1. Identity

Document	STAR-PKG-STAR-ELEMENTS-EN · v2.1 · 2026-11-01
Package	star-elements
Version	0.2.4
Capability	star.orbit.anomalies
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_elements

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_elements-0.2.4-py3-none-any.whl
```

FILE	SHA-256
star_elements-0.2.4-py3-none-any.whl	4454a65500bed1ed62637c8715b953b748a3f1a1a7f1c19b4f79b0334e3fed3b
star_elements.py	7b54ae0a1b1796edbc6cc38206d431dec18e3024b95a99fa76c3846e98002044
star_elements-0.2.4-source.zip	9832dce55a89f3e841c4eef7892bfc2eec65d9b47c3b87fe6d6ee08f51d7b943

5. What it does, and its limits

Classical orbital elements to and from Cartesian state. Standard library only.

Hyperbolic/parabolic anomalies, equinoctial or other non-singular element sets (needed for equatorial/circular orbits), mean elements, any flight use.

- `rv_to_coe(r, v, mu=398600.4418) → (h, e, i, raan, argp, nu)` (Curtis Algorithm 4.2); units km, km/s, radians.
- `coe_to_rv(h, e, i, raan, argp, nu, mu) → (r, v)` (Curtis Algorithm 4.5).
- Singular or invalid cases raise `ValueError`: equatorial (no RAAN), circular (no argp), rectilinear, h <= 0, hyperbolic true anomaly beyond the asymptotes. No arbitrary angle is ever returned for an undefined element.
- (0.2.0) Anomalies for 0 <= e < 1: `mean_to_eccentric` (Kepler, Newton; RuntimeError if not converged), `eccentric_to_true`, `true_to_eccentric`, `eccentric_to_mean`; e >= 1 raises `ValueError`.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_elements_0_2_4,
  author  = {Cavazzini, Andrea},
  title   = {{star-elements}: Classical orbital elements to and from Cartesian state (Curtis Algorithms 4.2 / 4.5)},
  version = {0.2.4},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_elements/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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